COMPUTER ENGINEERING

QUEZON CITY

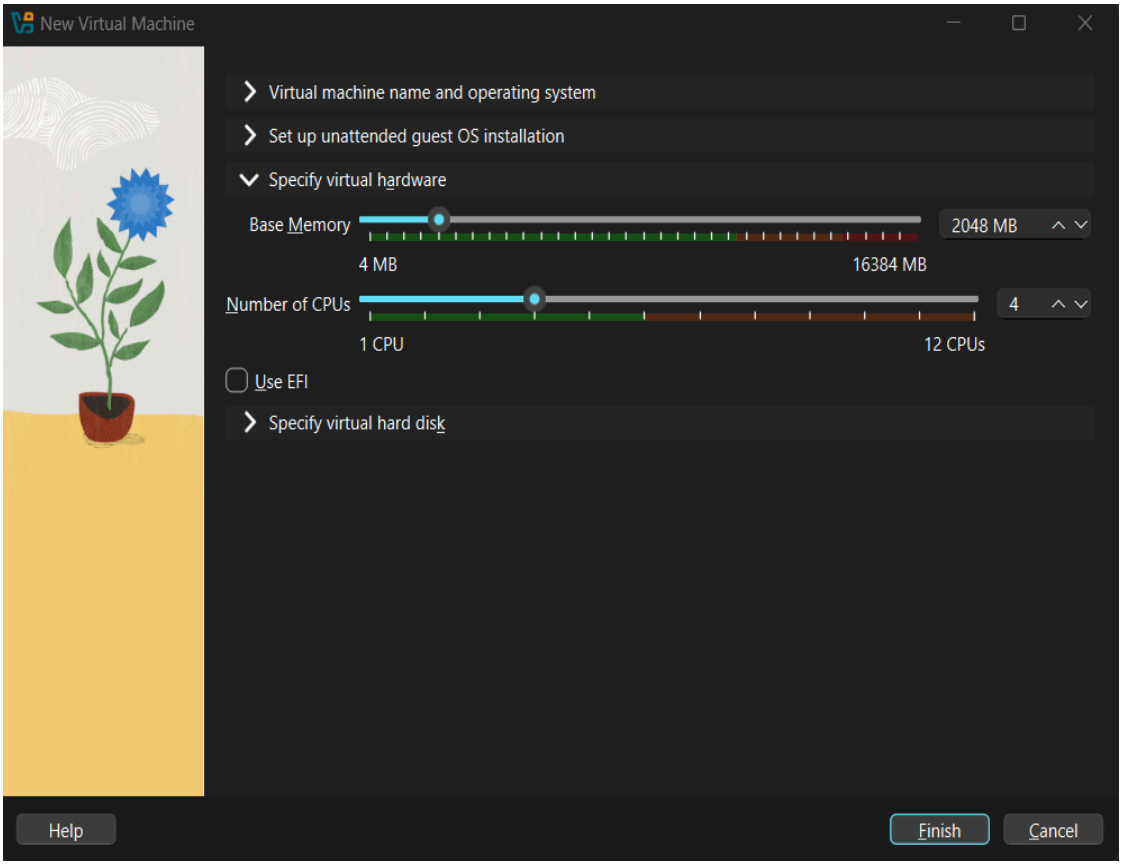
COURSE CODE/TITLE:	CPE212 - Automatic Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	23/07/2026	**DATE SUBMITTED:**	23/07/2026
NAME:	Que, Josef Leroy P.	**INSTRUCTOR:**	Engr. Jimlord Quejado

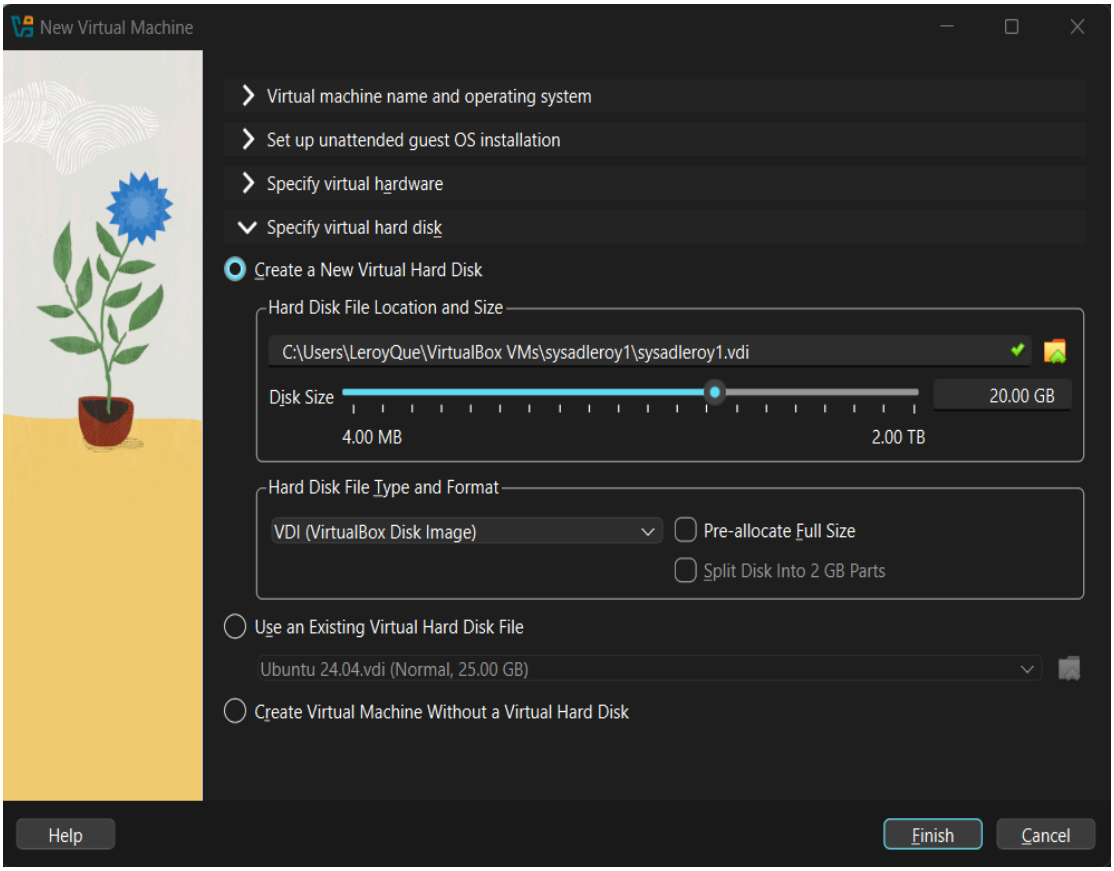
ACTIVITY NO. 3.1 **Install SSH Server on CentOS or RHEL 8**

1. PROCEDURE OUTPUT

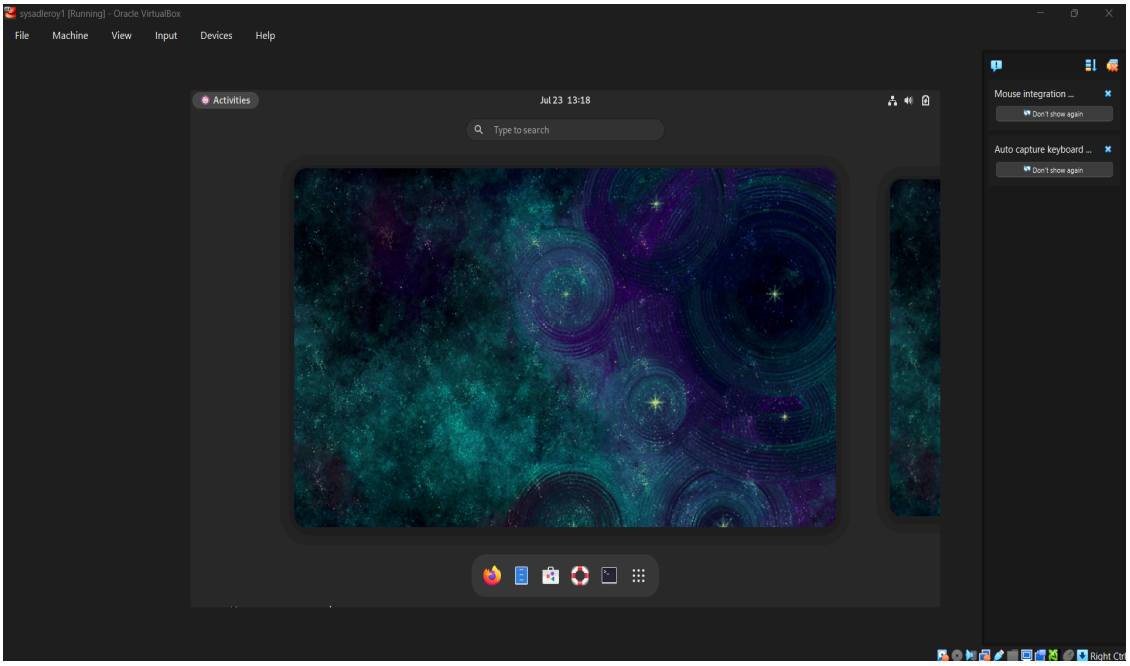
Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.





3. Install the downloaded image.
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package *openssh*

1. Install the ssh server package *openssh* by using the *dnf* command:

$ dnf install openssh-server

```
[root@localhost sysadleroy1]# dnf install openssh-server
CentOS Stream 9 - BaseOS                        1.7 MB/s | 9.0 MB     00:05
CentOS Stream 9 - AppStream                     4.1 MB/s |  28 MB     00:06
CentOS Stream 9 - Extras packages                33 kB/s |  21 kB     00:00     A
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost sysadleroy1]# S
```

2. Start the *sshd* daemon and set to start after reboot:

$ systemctl start sshd
$ systemctl enable sshd

```
[root@localhost sysadleroy1]# systemctl start sshd
[root@localhost sysadleroy1]# systemctl enable sshd
```

3. Confirm that the sshd daemon is up and running:

$ systemctl status sshd

```
[root@localhost sysadleroy1]# systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-07-23 13:17:47 PST; 4min 34s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 973 (sshd)
      Tasks: 1 (limit: 10496)
     Memory: 780.0K (peak: 1.8M)
        CPU: 12ms
     CGroup: /system.slice/sshd.service
             └─973 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 13:17:47 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jul 23 13:17:47 localhost.localdomain sshd[973]: Server listening on 0.0.0.0 po>
Jul 23 13:17:47 localhost.localdomain sshd[973]: Server listening on :: port 22.
Jul 23 13:17:47 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```

4. Open the SSH port 22 to allow incoming traffic:

$ firewall-cmd --zone=public --permanent --add-service=ssh
$ firewall-cmd --reload

```
[root@localhost sysadleroy1]# firewall-cmd --zone=public --permanent --add-servi
ce=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@localhost sysadleroy1]# firewall-cmd --reload
success
```

5. Locate the ssh server man config file */etc/ssh/sshd_config* and perform custom configuration. Every time you make any change to the */etc/ssh/sshd-config* configuration file reload the *sshd* service to apply changes:

```
# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

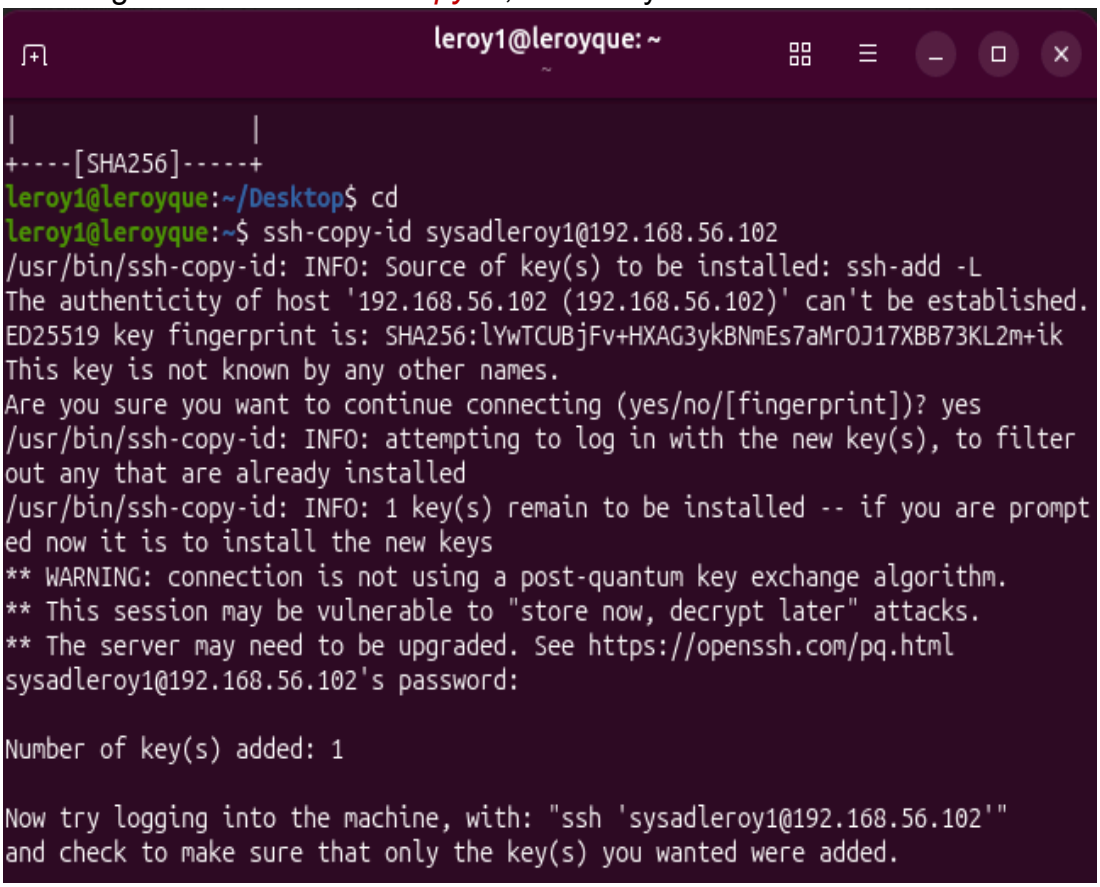
Analysis: PermitRootLogin should have ‘no’.

$ systemctl reload sshd

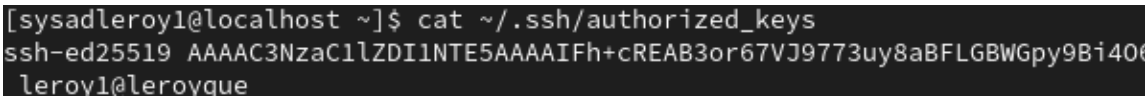
```
[root@localhost sysadleroy1]# nano /etc/ssh/sshd_config
[root@localhost sysadleroy1]# systemctl reload sshd
```

Task 3: Copy the Public Key to CentOS

1. Make sure that *ssh* is installed on the local machine.
2. Using the command *ssh-copy-id*, connect your local machine to CentOS.



3. On CentOS, verify that you have the *authorized_keys*.



Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.

Show evidence that you are connected.

```
leroy1@leroyque:~$ ssh sysadleroy1@192.168.56.102
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 13:17:56 2026
[sysadleroy1@localhost ~]$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:
Answer the following:
1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?
   - While making a choice between Debian and Red Hat, the decision depends upon the needs of the user in terms of how much support is needed from the software provider. When selecting the right option, it is also important to consider the commands that the user is comfortable with.
2. What are the main differences between Debian and Red Hat Linux distributions?
   - Debian is popular for regular use by users working on projects, while Red Hat is meant for larger enterprises and security structure and control. The two different software solutions employ different packages with Debian using deb packages and apt for executing commands, while Red Hat incorporates rpm packages and dnf commands. The differences in security policy are evident where Debian implements AppArmor and Red Hat is reliant on SELinux rationale.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Idea formatting**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was used solely to structure my ideas in an organized manner.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.



I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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